Simplifying Expressions

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 83 minutes · Total marks: 83

KEY RULES

Collect like terms	Add / subtract terms with the same letters and powers
Multiply	$3a \times 4a^2 = 12a^3$ (multiply numbers, add powers)
Divide	Cancel numbers and subtract powers
Index laws	$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$

WORKED EXAMPLE 1

[2 marks]

Simplify $3a + 5b - a + 2b$

Step 1	Collect the a terms	$3a - a = 2a$
Step 2	Collect the b terms	$5b + 2b = 7b$

Answer: $2a + 7b$

WORKED EXAMPLE 2

[2 marks]

Simplify $4x^2y \times 3xy^3$

Step 1	Multiply the numbers	$4 \times 3 = 12$
Step 2	Add the powers of x	$x^2 \times x = x^3$
Step 3	Add the powers of y	$y \times y^3 = y^4$

Answer: $12x^3y^4$

WORKED EXAMPLE 3

[3 marks]

Simplify $\dfrac{6x^5y^2}{3x^2y}$

Step 1	Divide the numbers	$6 \div 3 = 2$
Step 2	Subtract the powers of x	$x^{5-2} = x^3$
Step 3	Subtract the powers of y	$y^{2-1} = y$

Answer: $2x^3y$

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Simplifying Expressions

1 (a) Simplify $e^8 \div e^2$

.....

(1)

b) Expand and simplify $(x - 3)(x + 1)$

.....

(2)

(Total for Question 1 is 3 marks)

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2 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms

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3 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-}$

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11 Express $\frac{5}{3} - \frac{x+2}{2x}$ as a single fraction in its simplest terms.

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5

Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form

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6 Expand and simplify $3x(2x+5)(7x-4)$

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7 Expand and simplify $3x(2x-5)$

Show clear algebraic working.

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8 Expand and simplify $4x(3x+1)(2x-3)$

Show your working clearly.

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9 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

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10 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

.....

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11 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-2}$

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12 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

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13 Simplify fully $\dfrac{4x^2 - 9}{2x^2 - 5x - 12}$

Show clear algebraic working.

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14 Simplify fully $\dfrac{6x^3 + 13x^2 - 5x}{4x^2 - 25}$

.....

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15 Express $\left(\dfrac{20}{x^2 - 36} - \dfrac{2}{x - 6}\right) \times \dfrac{1}{4 - x}$ as a single fraction in its simplest form.

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16 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

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17 Show that $\dfrac{16x^2 - 36}{x - 7} \div \dfrac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

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18 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

.....

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19 Write $4-\left[(3x-5)\div\frac{3x^2+x-10}{4x-1}\right]$ as a single fraction in its simplest form.

.....

(Total for Question 26 is 4 marks)

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CHALLENGE QUESTIONS

20 (a) Simplify $m^5 \times m^2$

.....

(1)

(b) Simplify $c^{11} \div c^3$

.....

(1)

(c) Simplify $(a^5)^3$

.....

(1)

(d) Expand and simplify $4(2x + 3) + 2(x + 5)$

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 5

21 **23** $\dfrac{4x^2-4x-120}{5x^2-180} \div \dfrac{x^2+5x}{10x^2+60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ .....

(Total for Question 23 is 4 marks)

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22 Simplify $\left(x^2 - 4\right) \div \left(\dfrac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\dfrac{ax^2}{bx + c}$ where a, b and c are integers

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23 Write

$$\frac{4x^2 - 17x - 15}{2x - 1} \times \frac{2x^2 - 7x + 3}{x^2 - 25} + (29 - 4x)$$

as a single fraction in its simplest form.

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24 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

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TOTAL FOR PAPER IS 83 MARKS

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